Log Variable Reference

> **Source**: 115 log files, 82,238 total rows
> **Experiment**: Continuous Recognition Memory Experiment

File Format

Each `.log` file is a **CSV** with 14 columns. The first row is the header:

```
participant_ID, Timestamp, Event, Stimulus, isTarget, isValidation,
isRepeat, Button, Accuracy IR, Accuracy WR, Reaction_time_IR,
Reaction_time_WR, CR_IR, CR_WR
```

Variables

1. `participant_ID`

Unique numeric ID assigned to each participant by the server (`/api/participants/next`).

State	Description
`<integer>` (e.g. `232` , `400`)	Server-assigned participant number; consistent within a single log file
`N/A`	Fallback if the server did not return an ID (not observed in these logs)

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2. `Timestamp`

Unix epoch timestamp in **milliseconds** (`Date.now()`) marking when the log entry was created.

State	Description
`<unix_ms>` (e.g. `1770218862295`)	Milliseconds since January 1, 1970 UTC

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3. Event

The type of experimental event being logged — the primary key for interpreting each row.

State	Description
`Practice Sentence shown`	A sentence was displayed during the **practice** block
`Practice IR pressed`	Participant pressed Spacebar (item recognition) during **practice**
`Practice Missed`	Participant failed to press Spacebar on a practice repeat (timed out at 4 s)
`Practice WR pressed`	Participant responded Yes/No on the word-recognition question during **practice**
`Sentence shown`	A sentence was displayed during the **main** experiment
`IR pressed`	Participant pressed Spacebar (item recognition) during the **main** experiment
`WR pressed`	Participant responded Yes/No on the word-recognition question in the **main** experiment
`Validation IR pressed`	Spacebar pressed on a **validation-repeat** filler — correct detection
`Validation Wrong IR pressed`	Spacebar pressed on a **non-repeat** trial — false alarm tracked for validation
`Validation Missed`	Participant did **not** press Spacebar on a validation repeat (timed out)
`gap_time`	Logs the duration of the fixation-cross gap between trials
`Rest Phase started`	A rest/break period began between experiment blocks

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4. Stimulus

The sentence ID or form identifier associated with the event.

State	Description
Practice IDs	
`HH_<N>_A` / `HH_<N>_P`	Practice target sentence (High-High category, Active/Passive voice)
`HL_<N>_A`	Practice filler sentence (High-Low category)
Main Experiment IDs	

State	Description
`HVL <N> A` / `HVL_<N>_P`	Target — High Verbal memorability, Low non-verbal memorability (Active/Passive)
`LVH <N> A` / `LVH_<N>_P`	Target — Low Verbal memorability, High non-verbal memorability
`LVL <N> A` / `LVL_<N>_P`	Target — Low Verbal memorability, Low non-verbal memorability
`HF_<N>_A`	Filler sentence in the main experiment
`N/A`	Non-stimulus events (`gap_time` , `Rest Phase started`)

Voice suffixes: `_A` = Active voice, `_P` = Passive voice.

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5. `isTarget`

Whether the current trial is a **target** trial (has a recognition condition), as opposed to a filler or validation.

State	Description
`true`	Trial has a non-null condition that is not `"validation_repeat"` — it is a target sentence
(empty)	Trial is a filler or a validation repeat

Derived in code as: `currentTrial.condition !== null && currentTrial.condition !== "validation_repeat"`

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6. `isValidation`

Whether the trial is a **validation repeat** — a filler sentence re-presented nearby to check participant attentiveness.

State	Description
`true`	Trial's condition is `"validation_repeat"`
(empty)	Not a validation trial (target, filler first-presentation, or non-trial event)

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7. `isRepeat`

Whether this is a **second (or later) presentation** of a sentence — the recognition probe.

State	Description
`true`	This sentence was shown earlier and is now being re-presented for recognition
(empty)	First presentation of the sentence

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8. `Button`

The button/input the participant pressed, or metadata about the event.

State	Description
`N/A`	No button press (e.g. `Sentence shown` events)
`Spacebar`	Participant pressed Space to indicate recognition (IR response)
`Yes`	Participant pressed the `A` key → "Yes, the exact same words" (WR response)
`No`	Participant pressed the `D` key → "No, different words" (WR response)
`None`	Participant did not respond before timeout (Missed events)
`Rest`	Rest/break phase marker
`GapTime_<N>`	Records the fixation-cross gap duration in ms (e.g. `GapTime_1095` = 1095 ms)

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9. `Accuracy_IR` (Item Recognition Accuracy)

Whether the participant **correctly** identified the sentence as old (repeat) or new (first presentation).

State	Description
`1`	**Hit** — participant pressed Spacebar on a repeat trial (`is_repeat === true`)
`0`	**False alarm** — participant pressed Spacebar on a first-presentation trial; OR **Miss** — participant failed to press on a repeat (timeout)
`N/A`	Not applicable (sentence display, WR response, gap, consent, demographics events)

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10. `Accuracy_WR` (Word Recognition Accuracy)

Whether the participant correctly identified if the **exact same wording** or a voice-transformed version was used.

State	Description
`1`	**Correct** — pressed "Yes" on a `WR` condition (same words repeated) OR pressed "No" on a `Transformed` condition (active↔passive voice change)
`0`	**Incorrect** — wrong button for the given condition
`N/A`	Not applicable (non-WR events, or trials where no WR correct answer exists)
(null/empty)	Condition is neither `WR` nor `Transformed` (fillers, validation repeats) — no ground truth to score against

Scoring logic:

- **WR condition** (same words): correct answer is `Yes` → `accuracyWR = (choice === "Yes") ? 1 : 0`
- **Transformed condition** (active↔passive flip): correct answer is `No` → `accuracyWR = (choice === "No") ? 1 : 0`

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11. `Reaction_time_IR` (ms)

Time from **sentence onset** to **Spacebar press** (item-recognition reaction time).

State	Description
`<integer_ms>` (e.g. `1069`, `2812`)	Milliseconds from `displayStartTime` to the Spacebar press
`N/A`	No IR response occurred (sentence display, WR, gap, missed, non-response events)

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12. `Reaction_time_WR` (ms)

Time from **IR response** to **Yes/No button press** (word-recognition reaction time).

State	Description
`<integer_ms>` (e.g. `318`, `5086`)	Milliseconds from the IR Spacebar press to the Yes (`A`) or No (`D`) key press
`N/A`	No WR response (non-WR events, sentence display, IR events, gaps, etc.)

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13. No need to log `CR_IR` and `CR_WR` (condition labels) since they can be derived from the stimulus ID and trial structure. These columns are currently unused in the logs.

Trial Types Summary

Trial Type	`isTarget`	`isValidation`	`isRepeat`	`Accuracy IR` scored?	`Accuracy WR` scored?
Filler (first presentation)	*(empty)*	*(empty)*	*(empty)*	Only if Spacebar pressed (false alarm = 0)	No (N/A)
Target (first presentation)	`true`	*(empty)*	*(empty)*	Only if Spacebar pressed (false alarm = 0)	No (N/A)
Target repeat (WR condition)	`true`	*(empty)*	`true`	Yes (hit = 1)	Yes (Yes = 1, No = 0)
Target repeat (Transformed condition)	`true`	*(empty)*	`true`	Yes (hit = 1)	Yes (No = 1, Yes = 0)
Validation repeat	*(empty)*	`true`	`true`	Yes (hit = 1)	No (N/A)